

Smart Helmet for Cyclists and Motor Cyclists

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Member Functions

1. Health Monitoring System

The helmet continuously monitors vital signs such as heart rate and blood pressure throughout the user's journey. If abnormal readings are detected such as signs of a potential heart problem or heart attack the system immediately alerts the user with a voice warning and activates hands-free voice commands. For example, the user can simply say “Call Contact” or “Send Help” to notify emergency contacts without needing to touch the device. In less critical cases, the mobile app analyzes current health trends and journey data to offer smart advice, such as recommending a short break “You have 50 miles ahead, but based on your heart rate, it's better to rest now your estimated arrival time remains unaffected.” This feature not only supports emergency situations but also helps prevent them by promoting healthy travel habits. The app further encourages user engagement and motivation through personalized feedback and positive reinforcement, making long journeys safer and more manageable.

Technologies

- Heart rate sensor - like MAX30100 or MAX30102
- Blood pressure sensor - like SEN-04277
- Temperature sensor - like LM35 or DS18B20
- ESP32 - to read sensor data, process info, and connect to the mobile app
- Voice recognition - using Pico voice Porcupine for offline commands

2. Emotion and Stress Monitoring with Real-Time Feedback

Throughout the journey, the helmet continuously monitors the user's emotional state using an EEG module connected via Bluetooth Low Energy (BLE). The brainwave data is sent to a mobile app, where a built-in stress classification model analyzes patterns to detect stress levels. If high stress is detected, the app provides instant recommendations to help the user relax such as suggesting a short break, offering breathing exercises, or

playing calming music to reduce tension. The user can view their stress status in real time, with alerts like “High Stress Detected consider resting before continuing.” This feature promotes mental well-being during rides and adds a layer of emotional support, making the helmet not just a safety tool, but also a mental health companion.

Technologies

- EEG Module - Detects brainwave activity for stress detection
- Bluetooth Low Energy (BLE) - Sends EEG data from the headset to the mobile app wirelessly

3. Post-Journey Risk Assessment and Report Generation

At the end of each journey, the helmet generates a personalized risk report based on realtime sensor data collected during the ride. The system uses a combining accelerometer, gyroscope, and magnetometer to detect risky speed changes, sudden movements, sharp turns, and potential fall directions. Additionally, it evaluates stress levels recorded through EEG sensors to identify moments of emotional strain. All this data is analyzed and summarized into an easy-to-read report in the mobile app, showing sections like “High-Speed Risk Zones”, “Stress Peaks” and “Potential Danger Moments.” For example, if the user rode at high speeds while under stress or during sharp turns, the report highlights this as a high-risk behavior. This function helps users understand their riding patterns, improve future safety, and keep a log of their physical and emotional conditions throughout the journey.

Technologies

- Accelerometer - for speed and sudden movement
- Gyroscope - for rotation and balance
- EEG Module - Monitors brain activity to detect stress during journey •
Microcontroller - Reads IMU and EEG data, sends to app

4. Predictive Mental Health and Fatigue Monitoring (Enhanced with Heart Rate Analysis)

This function continuously monitors the user's brainwave patterns using an EEG sensor and combines them with real-time heart rate data to detect signs of long-term stress, fatigue, and early mental health risks. By analyzing trends in alpha, beta, and theta brainwave activity alongside heart rate variability (HRV) and pulse rate, the system can identify potential conditions such as burnout, chronic fatigue, anxiety, or depression before they escalate.

A machine learning model processes both EEG and heart rate data to detect abnormal physiological patterns that indicate mental or emotional strain. When risk levels are detected, the connected mobile app issues early alerts, offering personalized self-care recommendations such as resting, reducing cognitive workload, or seeking medical advice.

In more critical or persistent cases—where both EEG and heart signals suggest abnormal stress or fatigue—the system can recommend a medical check-up or alert the user to consult a mental health professional. This integrated monitoring approach provides a proactive, multi-modal assessment that not only enhances road safety for riders but also promotes long-term mental and emotional well-being.

In addition, the app offers educational resources about mental fatigue, stress management techniques, and motivational content to support healthier daily routines—especially for individuals undergoing extended travel or operating in high-stress environments.

Technologies

- EEG Module – Captures brainwave activity (Alpha, Beta, Theta, Delta waves)
- Heart Rate Sensor – Monitors pulse and heart rate variability (HR/HRV) in real-time
- Microcontroller – Collects and transmits EEG and heart rate data to the mobile application
- Machine Learning Model – Predicts potential fatigue or mental health risks using combined EEG and HR features
- Alert & Recommendation System – Sends early warnings and guides users with evidence-based self-care and medical suggestions